SimCity: A Unified Neural-Epidemiological Architecture for Modelling Viral Information Dynamics on Multiplex Internet Graphs

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Abstract

The dynamics of internet virality exhibit both macroscopic epidemiological patterns and microscopic, graph-structural volatility. Current literature largely isolates these domains: macroscopic models (e.g., SEIR) fail to capture the transient topology of social networks, while microscopic graph models (e.g., Temporal Graph Networks) lack population-level infection dynamics. In this paper, we present **SimCity**, a unified neural-epidemiological architecture that bridges this gap. We introduce a Stochastic SEIR-Z-D compartmental model grounded by a Degree-Correlated Heterogeneous Mean-Field (HMF) approximation, which maps continuous-time Temporal Graph Network (TGN) embeddings directly into epidemiological transition rates. We couple this with a Neural Multivariate Hawkes Process for cross-platform narrative transfer and a Filippov-compliant smooth Deffuant model for agent behavioural evolution. Empirically, we make three contributions: (1) we show that raw engagement-count regression — the de-facto virality metric — is intrinsically noise-dominated under near-critical cascade dynamics (even an oracle handed the true excitation state attains negative residual skill), arguing for timing- and transfer-based evaluation; (2) on a controlled multiplex benchmark with ground-truth cross-platform excitation, our narrative-conditioned intensity model detects *which* narratives will transfer across platforms (AUC 0.653 ± 0.005 over three seeds, $p < 1.1 \times 10^{-4}$, robust to popularity and reach confounds) — a task on which any static Hawkes process is random by construction; and (3) the result transfers to the wild: on a 32k-event live Hacker News + GDELT corpus from our released accumulating collector, we diagnose a sign non-identifiability of the narrative-conditioned excitation head (seeds converge to the same ranking up to mirror image, $\rho = 0.96$), resolve it with leakage-free validation-split orientation, and obtain AUC 0.778 ± 0.104 over three seeds (every seed $p \leq 1.2 \times 10^{-8}$), against an anti-predictive popularity baseline.

1 Related Work

Temporal graph representation learning. Continuous-time dynamic graph models embed evolving interaction streams: JODIE (Kumar et al., 2019) learns coupled user/item trajectories, DyRep (Trivedi et al., 2019) drives embeddings with a two-time-scale point process, TGAT (Xu et al., 2020) introduces functional time encoding for temporal attention, and TGN (Rossi et al., 2020) unifies these with a memory module, which we adopt as our structural backbone. These models are trained for link-level prediction; they contain no population-level infection dynamics and no mechanism for platform-to-platform transfer, which is precisely the gap SimCity's HMF bridge (section 4) and cross-platform Hawkes head (section 5) address.

Hawkes processes and neural temporal point processes. Self- and mutually-exciting processes (Hawkes, 1971) underpin much of social-media modelling. Neural parameterisations replace fixed kernels with learned dynamics: RMTPP (Du et al., 2016), the Neural Hawkes Process (Mei and Eisner, 2017), and the Transformer Hawkes Process (Zuo et al., 2020). Closer to our setting, NETRATE (Gomez-Rodriguez et al., 2011) infers diffusion networks from cascades, COEVOLVE (Farajtabar et al., 2017) couples diffusion with network growth, and prior work separates endogenous from exogenous influence (Myers et al., 2012) — our GDELT-driven $\mu_i(t)$ follows this exogenous/endogenous decomposition. Our contribution is orthogonal to kernel expressiveness: we condition the *cross-platform excitation matrix itself* on evolving graph and narrative state, which is what enables per-narrative transfer ranking (section 8.3) — a static or globally-parameterised Hawkes model is random at that task by construction.

Cascade and popularity prediction. Feature-based and deep models predict cascade size: cascade growth was framed as a prediction problem in (Cheng et al., 2014), SEISMIC (Zhao et al., 2015) and Hawkes intensity processes (Rizoiu et al., 2017; Kobayashi and Lambiotte, 2016) model retweet dynamics, reinforced Poisson processes model item popularity (Shen et al., 2014), and DeepCas/DeepHawkes (Li et al., 2017; Cao et al., 2017) learn end-to-end representations; community structure also signals virality (Weng et al., 2013). Our negative result (section 8.3) complements this line: under near-critical branching, next-window *counts* are noise-dominated even for an oracle, motivating timing- and transfer-based evaluation instead of raw magnitude regression.

Epidemic models on networks. Degree-based mean-field theory characterises epidemic thresholds on heterogeneous graphs (Pastor-Satorras and Vespignani, 2001; Boguñá and Pastor-Satorras, 2002; Pastor-Satorras et al., 2015). We import the degree-correlated HMF closure into a learned setting, using TGN embeddings to supply node-level infectivities that are aggregated into macroscopic SEIR-Z-D rates — to our knowledge the first bridge from continuous-time graph memory states to compartmental transition rates.

Cross-platform information flow. Empirical work documents inter-platform influence: the "web centipede" (Zannettou et al., 2017) traces news trajectories across communities, coordinated cross-platform disinformation campaigns have been analysed in (Wilson and Starbird, 2020), platform migrations reshape community behaviour (Horta Ribeiro et al., 2021), and false news spreads with distinctive dynamics (Vosoughi et al., 2018). These studies are measurement-focused; SimCity provides the corresponding *predictive* instrument — a causal, per-narrative transfer score evaluated prospectively on live Hacker News + GDELT data.

Opinion dynamics. Bounded-confidence models (Deffuant et al., 2000; Hegselmann and Krause, 2002) and their statistical-physics treatments (Castellano et al., 2009) evolve continuous opinions under confidence thresholds. We contribute a Filippov-compliant smooth relaxation (section 6) whose radicalization state feeds back into representation learning, coupling behavioural and structural dynamics.

2 Information Propagation: Stochastic SEIR-Z-D Model

To rigorously capture the volatile, non-deterministic nature of viral information spread, we formulate a continuous-time Stochastic Master Equation (SME). We introduce a **Zombie** (Z) state to model algorithmic resurfacing, and an **Archived/Dead** (D) absorbing state to ensure population conservation and a well-defined long-run limit.

2.1 State Definitions and Population Conservation

Let the network state at time t be defined by the discrete random variable vector

$$\mathbf{n}(t) = (n_S, n_E, n_I, n_R, n_Z, n_D).$$

The six compartments are:

- S (Susceptible): users who have not yet encountered the content.
- E (Exposed): users who have seen the content but not yet shared it.
- I (Infected): users actively sharing and propagating the content.
- R (Recovered): users who have lost interest (decayed engagement).
- Z (Zombie): content resurfaced from R via algorithmic recommendation or external shock.
- D (Archived/Dead): content in a final absorbing state; no further transitions are possible.

We explicitly enforce a *constant* total population:

$$N = n_S(t)+n_E(t)+n_I(t)+n_R(t)+n_Z(t)+n_D(t) \quad \forall t. \tag{1}$$

Remark 1. *On the internet N is not strictly constant (new accounts, lurkers becoming active). We treat N as fixed throughout a single cascade window, a standard simplification in compartmental internet models. Extensions to open populations are left as future work.*

2.2 Transition Rates and Jump Vectors

The system evolves via five discrete transitions. Each is characterised by a transition rate $W_k(\mathbf{n})$ and a state jump vector $\mathbf{r}_k \in \mathbb{Z}^6$.

T1. Exposure ($S \to E$): $\mathbf{r}_1 = (-1,\ +1,\ 0,\ 0,\ 0,\ 0)$

$$W_1(\mathbf{n}) \;=\; \beta(t)\cdot\frac{n_S\big(n_I+\theta\, n_Z\big)}{N}, \qquad (2)$$

where $\theta \geq 0$ scales the algorithmic boost of resurfaced zombie content.

T2. Activation ($E \to I$): $\mathbf{r}_2 = (0,\ -1,\ +1,\ 0,\ 0,\ 0)$

$$W_2(\mathbf{n}) \;=\; \sigma\, n_E. \qquad (3)$$

T3. Decay ($I \to R$): $\mathbf{r}_3 = (0,\ 0,\ -1,\ +1,\ 0,\ 0)$

$$W_3(\mathbf{n}) \;=\; \gamma_I\, n_I. \qquad (4)$$

T4. Resurgence ($R \to Z$): $\mathbf{r}_4 = (0,\ 0,\ 0,\ -1,\ +1,\ 0)$

$$W_4(\mathbf{n}) \;=\; \zeta(t)\, n_R, \qquad (5)$$

where $\zeta(t)$ is driven by cross-platform narrative shocks; see section 5 for the formal link.

T5. Archival ($Z \to D$): $\mathbf{r}_5 = (0,\ 0,\ 0,\ 0,\ -1,\ +1)$

$$W_5(\mathbf{n}) \;=\; \delta_D\, n_Z. \qquad (6)$$

D is absorbing: no outgoing transitions exist, closing the $Z \to R$ infinite-loop present in standard SEIR-Z formulations.

Remark 2. *The decay parameter in **T3** is denoted γ_I to avoid confusion with the Hawkes decay parameter $\hat{\gamma}_{ij}$ introduced in section 5.*

2.3 The Master Equation

The time evolution of the joint probability distribution $P(\mathbf{n}, t)$ is governed by:

$$\frac{dP(\mathbf{n},t)}{dt} \;=\; \sum_{k=1}^{5}\Big[W_k(\mathbf{n}-\mathbf{r}_k)\,P(\mathbf{n}-\mathbf{r}_k,\,t) \;-\; W_k(\mathbf{n})\,P(\mathbf{n},\,t)\Big]. \qquad (7)$$

This formulation fully specifies the Markov process: each term represents probability *flowing into* state $\mathbf{n}$ from $\mathbf{n}-\mathbf{r}_k$ minus probability *leaving* $\mathbf{n}$ via transition k. Taking expectations over $P(\mathbf{n},t)$ recovers the macroscopic ODEs, while the second moment yields virality variance.

3 Micro-Dynamics: Temporal Graph Network Integration

The macroscopic rates $\beta(t)$ and $\zeta(t)$ are *not* static parameters; they are derived dynamically from the evolving micro-structure of the social graph via a Temporal Graph Network (TGN) augmented with a **Virality Prediction Head**.

3.1 Core TGN (Rossi et al., 2020)

For a node v involved in an interaction with node u at time t:

$$\text{Message:}\quad \mathbf{s}_v(t) = \mathrm{GRU}\big(\mathbf{s}_v(t^-),\ \bar{m}_v(t)\big)\,, \qquad (8)$$

$$\text{where}\quad m_v(t) = \mathrm{Msg}\big(\mathbf{s}_v(t^-),\ \mathbf{s}_u(t^-),\ \Delta t,\ e_{vu}(t)\big)\,,$$

$$\text{Embedding:}\quad \mathbf{h}_v(t) = \mathrm{attn}\big(\mathbf{s}_v(t),\ \{\mathbf{s}_u(t)\}_{u\in\mathcal{N}(v)}\big)\,. \qquad (9)$$

3.2 Contextual Content Embedding

Content c is not a static entity; its contextual embedding evolves as users interact with it. To prevent self-referential circularity, attention weights are computed strictly from *pre-update* historical embeddings $\mathbf{h}(t^-)$:

$$a_{u,c} \;=\; \frac{\exp\!\big(\mathbf{h}_u(t^-)^\top \mathbf{W}_a\, \mathbf{h}_c(t^-)\big)}{\sum\limits_{j\in\mathcal{N}_c(t)} \exp\!\big(\mathbf{h}_j(t^-)^\top \mathbf{W}_a\, \mathbf{h}_c(t^-)\big)}, \qquad (10)$$

$$\mathbf{h}_c(t) \;=\; \mathrm{GRU}\left(\mathbf{h}_c(t^-),\ \sum_{u\in\mathcal{N}_c(t)} a_{u,c}\,\mathbf{h}_u(t^-)\right). \qquad (11)$$

At $t=0$, content is initialised as

$$\mathbf{h}_c(0) \;=\; \mathbf{W}_{\mathrm{proj}}\cdot \mathrm{LLM\_Encode(text)}, \qquad (12)$$

where the LLM encoder is *frozen* to prevent catastrophic forgetting, and $\mathbf{W}_{\mathrm{proj}}$ is a learned linear projection mapping the LLM output dimension to the TGN hidden dimension d.

3.3 Virality Prediction Head

Node and content embeddings are mapped to predictive epidemiological and Hawkes parameters via localised MLPs with Softplus output activations (enforcing positivity):

$$\hat{\beta}_{v,c}(t) = \mathrm{Softplus}(\mathrm{MLP}_\beta(\mathbf{h}_v(t)\,\|\,\mathbf{h}_c(t)))\,, \qquad (13)$$

$$\hat{\alpha}_{ij}(t) = \mathrm{Softplus}\big(\mathrm{MLP}_\alpha\big(\mathbf{h}_{\mathcal{P}_i}(t)\,\|\,\mathbf{h}_{\mathcal{P}_j}(t)\,\|\,\mathbf{h}_c(t)\big)\big)\,, \qquad (14)$$

$$\hat{\gamma}_{ij}(t) = \mathrm{Softplus}\big(\mathrm{MLP}_\gamma\big(\mathbf{h}_{\mathcal{P}_i}(t)\,\|\,\mathbf{h}_{\mathcal{P}_j}(t)\,\|\,\mathbf{h}_c(t)\big)\big)\,, \qquad (15)$$

where $\mathbf{h}_{\mathcal{P}_i}(t)$ is the platform-level embedding defined in section 3.4, $\hat{\beta}_{v,c}(t)$ feeds the HMF bridge (section 4), and $\hat{\alpha}_{ij}(t)$, $\hat{\gamma}_{ij}(t)$ feed the Neural Hawkes process (section 5).

3.4 Platform-Level Embedding

To aggregate node-level embeddings into platform-level representations for the Hawkes infectivity MLPs, we use **Influence-Weighted Attention Pooling**. Simple mean pooling would wash out super-spreader signals in heavy-tailed graphs; instead, we weight by the dynamic influence score $I(v, t)$ (defined in section 7):

$$\begin{aligned} \mathbf{h}_{\mathcal{P}_i}(t) &= \sum_{v \in V_{\mathcal{P}_i}(t)} \alpha_v(t)\, \mathbf{h}_v(t), \\ \alpha_v(t) &= \frac{\exp\bigl(\tau \cdot I(v,t)\bigr)}{\sum_{u \in V_{\mathcal{P}_i}(t)} \exp\bigl(\tau \cdot I(u,t)\bigr)}, \end{aligned} \tag{16}$$

where τ is a learnable temperature parameter.

3.5 Joint Optimisation: Homoscedastic Multi-Task Loss

All three objectives are formulated as regression/continuous tasks to satisfy the Gaussian likelihood assumption underlying Kendall et al.'s homoscedastic uncertainty weighting (Kendall et al., 2018):

$$\begin{aligned} \mathcal{L} &= \frac{1}{2\sigma_1^2}\mathcal{L}_{\text{TGN}} + \frac{1}{2\sigma_2^2}\mathcal{L}_{\text{Hawkes}} \\ &\quad + \frac{1}{2\sigma_3^2}\mathcal{L}_{\text{Virality}} + \log(\sigma_1\sigma_2\sigma_3), \end{aligned} \tag{17}$$

where $\sigma_1, \sigma_2, \sigma_3 > 0$ are learnable noise parameters.

(1) TGN Structural Loss — InfoNCE. To learn embeddings that discriminate temporal edges from noise, we use a temporal contrastive loss with degree-proportional negative sampling $P_n(v) \propto d_v$ (uniform sampling produces trivially easy negatives in heavy-tailed graphs): Writing $s_t(u,v) := \mathbf{h}_u(t)^\top \mathbf{h}_v(t)$ for the temporal edge score,

$$\begin{aligned} \mathcal{L}_{\text{TGN}} = -\sum_{(u,v,t)\in\mathcal{E}} \Bigl[s_t(u,v) \\ - \log\Bigl(e^{s_t(u,v)} + \sum_{v_\neg \sim P_n} e^{s_t(u,v_\neg)}\Bigr)\Bigr]. \end{aligned} \tag{18}$$

We draw $M = 20$ negatives per positive edge, following Rossi et al. (2020) as a baseline; a systematic ablation over $M \in [10, 200]$ is left as future work.

(2) Hawkes Negative Log-Likelihood.

$$\mathcal{L}_{\text{Hawkes}} = -\sum_{i=1}^{|\mathcal{P}|} \left[\sum_{t_k^{(i)}} \log \lambda_i\Bigl(t_k^{(i)}\Bigr) - \int_0^T \lambda_i(t)\, dt \right]. \tag{19}$$

The compensator integral is tractable via the piecewise-recursive update described in section 5.3.

(3) Virality Regression Loss.

$$\mathcal{L}_{\text{Virality}} = \frac{1}{|\mathcal{C}|} \sum_{c\in\mathcal{C}} \Bigl(\log E_c - \log \hat{E}_c\Bigr)^2, \tag{20}$$

where E_c is the true cumulative engagement count and $\hat{E}_c$ is the model's prediction.

4 The Micro-Macro Bridge: Degree-Correlated HMF

The individual node-level predictions $\hat{\beta}_{v,c}(t)$ must be aggregated into the macroscopic SEIR-Z-D exposure rate $\beta(t)$. A naive homogeneous mean-field approximation assumes uncorrelated, exchangeable nodes — a catastrophic assumption for the highly assortative, heavy-tailed graphs characteristic of social media.

4.1 Full Degree-Correlated HMF Formulation

Let $P(k'|k)$ be the conditional probability that a node of degree k connects to a node of degree k'. Stratifying infected nodes by degree and weighting by the source degree k' (the number of susceptible contacts an infected node of degree k' can reach):

$$\beta(t) = \sum_k \sum_{k'} P(k'|k) \cdot k' \cdot \mathbb{E}_{v\in\mathcal{I}_{k'}(t)}\Bigl[\hat{\beta}_{v,c}(t)\Bigr]. \tag{21}$$

Assumption 1 (Stationarity of Mixing Structure)**.** *To avoid the $\mathcal{O}(V^2)$ cost of recomputing $P(k'|k)$ at every event, we estimate it as a static approximation over the full observation window. This is justified because while individual node degrees $k_v(t)$ are highly dynamic, the macroscopic assortative mixing structure of a given platform remains approximately stationary over the timescale of a single cascade.*

4.2 Effective Temporal Degree

Because normalised TGN attention weights sum to 1 by construction (softmax), they cannot serve as a proxy for graph volume. Instead, we define the effective temporal degree $k_v(t)$ directly from the event stream within the TGN's explicit memory horizon ΔT_{mem}:

$$k_v(t) = \big|\{u \in \mathcal{V} \mid \exists (v,u,\tau) \text{ with } t - \Delta T_{\text{mem}} \le \tau \le t\}\big|. \quad (22)$$

This guarantees that hubs yield large $k_v(t)$ and lurkers yield $k_v(t) \approx 0$, restoring physical validity to eq. (21).

5 Cross-Platform Transfer: Neural Multivariate Hawkes Process

Internet narratives migrate across platforms: a burst on Reddit triggers a cascade on Twitter/X, which bleeds into mainstream news (GDELT). We model this mutually exciting cross-platform transfer with a **Neural Multivariate Hawkes Process**.

5.1 Conditional Intensity Function

Let $N_i(t)$ be the counting process of narrative events on platform $i \in \{1, \ldots, |\mathcal{P}|\}$. The conditional intensity given history $\mathcal{H}_t$ is:

$$\lambda_i(t) = \mu_i(t) + \sum_{j=1}^{|\mathcal{P}|} \sum_{t_k^{(j)} < t} \hat{\alpha}_{ij}(t_k)\, e^{-\hat{\gamma}_{ij}(t_k)\left(t - t_k^{(j)}\right)}, \quad (23)$$

where:

- $\mu_i(t)$ is the exogenous background rate on platform i (see section 8.1 for its empirical definition);
- $\hat{\alpha}_{ij}(t_k) \ge 0$ is the dynamic infectivity of platform j on platform i at event time t_k, output by MLP_α (eq. (14));
- $\hat{\gamma}_{ij}(t_k) > 0$ is the dynamic exponential decay, output by MLP_γ (eq. (15)).

When $j = i$, $\hat{\alpha}_{ii}$ captures *intra-platform* self-excitation (e.g., Reddit posts triggering more Reddit posts); $j \neq i$ captures *cross-platform* narrative transfer.

Proposition 1 (Cascade Threshold). *Define the branching matrix $\mathbf{\Gamma}$ with entries $\Gamma_{ij} = \hat{\alpha}_{ij}/\hat{\gamma}_{ij}$. A narrative exhibits subcritical (fading) behaviour if the spectral radius $\rho(\mathbf{\Gamma}) < 1$. If $\rho(\mathbf{\Gamma}) \ge 1$, the narrative goes globally viral.*

Proof sketch. For fixed parameters, $\Gamma_{ij} = \int_0^\infty \hat{\alpha}_{ij} e^{-\hat{\gamma}_{ij} s}\, ds$ is the expected number of platform-i offspring of a platform-j event, so $\mathbf{\Gamma}$ is the mean offspring matrix of the associated multitype branching process; extinction (stationarity of the linear Hawkes process) holds iff $\rho(\mathbf{\Gamma}) < 1$ (Hawkes, 1971; Brémaud and Massoulié, 1996). In our model $\hat{\alpha}, \hat{\gamma}$ are piecewise-constant between events (section 5.3), so the criterion applies locally per inter-event interval; $\rho(\mathbf{\Gamma}(t))$ then acts as an instantaneous criticality index rather than a global stationarity guarantee. □

5.2 Linking $\zeta(t)$ to the Hawkes Process

The resurgence rate $\zeta(t)$ in the SME (eq. (5)) is formally defined as a function of the aggregate cross-platform Hawkes intensity on the content's originating platform i^*:

$$\zeta(t) = f(\lambda_{i^*}(t)) = \phi \cdot \max\big(0,\ \lambda_{i^*}(t) - \bar{\lambda}_{i^*}\big), \quad (24)$$

where $\bar{\lambda}_{i^*}$ is the rolling baseline intensity and $\phi > 0$ is a scaling hyperparameter. Surges above baseline directly elevate the probability of zombie resurgence.

5.3 Piecewise-Recursive Tractability of the Compensator

Standard Hawkes processes with static α_{ij}, γ_{ij} admit a single closed-form compensator. In our model, $\hat{\alpha}_{ij}(t_k)$ and $\hat{\gamma}_{ij}(t_k)$ are TGN outputs that update *discretely* at each event, remaining piecewise-constant between events. The exponential kernel then permits a recursive analytical update: between consecutive events t_k and t_{k+1}, the integral contribution

$$\int_{t_k}^{t_{k+1}} \lambda_i(t)\, dt$$

is closed-form given the fixed parameters at t_k, reducing the full compensator to a tractable sum over event intervals.

6 Agent Behavioural Policy: Smooth Deffuant Model

For simulating node-level opinion evolution, we adapt the continuous-opinion Deffuant model to (i) operate in continuous time and (ii) incorporate platform-specific confidence bounds that shrink with user radicalization.

6.1 Filippov-Compliant Smooth Formulation

The standard Deffuant model uses a hard indicator function $\mathbf{1}_{|x_v - x_c| \le \epsilon_p}$, which destroys Lipschitz continuity and violates the Picard–Lindelöf existence and uniqueness theorem. We replace it with a smooth logistic approximation:

$$\dot{x}_v(t) = \eta \cdot \big(x_c - x_v(t)\big) \cdot \sigma\Big(\rho \cdot \big(\epsilon_p(v) - |x_v(t) - x_c|\big)\Big), \tag{25}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$, η is the opinion convergence rate, and the steepness hyperparameter is bound as

$$\rho = \frac{\kappa}{\epsilon_{\text{base}}}, \tag{26}$$

with κ a fixed scaling constant. As $\rho \to \infty$, eq. (25) recovers the exact Deffuant threshold; finite ρ guarantees smoothness at the boundary. Appendix B discusses the expected insensitivity of steady-state opinion clusters for $\rho > 10^2$.

Remark 3. $x_c := x_u(t_{\text{post}})$*: content carries the immutable opinion state of its creator at the moment of posting.*

6.2 Dynamic Radicalization

Platform confidence bounds shrink as users become radicalized:

$$\epsilon_p(v) = \epsilon_{\text{base}} \cdot \big(1 - \text{Radicalization}(v, t)\big), \tag{27}$$

where $\text{Radicalization}(v, t)$ is a convex combination of structural isolation and historical opinion rigidity:

$$\text{Radicalization}(v,t) = \alpha_r \underbrace{\left(1 - \frac{|\partial\mathcal{N}_v \cap \mathcal{C}_v|}{|\mathcal{N}_v|}\right)}_{\text{structural isolation}} + (1 - \alpha_r) \underbrace{\left(\frac{\sum_{\tau \le t} \mathbf{1}_{\text{rejected}}}{\sum_{\tau \le t} \mathbf{1}_{\text{exposed}}}\right)}_{\text{historical rigidity}}, \tag{28}$$

with $\mathcal{C}_v$ denoting node v's detected community and $\partial\mathcal{N}_v$ its cross-community neighbourhood. Because $\text{Radicalization}(v, t) \in [0, 1]$ by construction, $\epsilon_p(v) \ge 0$ is always satisfied. This scalar is **concatenated** to the raw node feature vector $\mathbf{x}_v(t)$ at every TGN timestep, permanently coupling behavioural polarisation with structural embedding generation.

7 Dynamic Influence Score and Super-Spreader Identification

Internet influence is highly transient; static topology metrics such as standard PageRank are insufficient. We define a **Dynamic Influence Score**:

$$I(v,t) = \omega_1 \cdot \text{TPR}(v,t) + \omega_2 \cdot \frac{dE(v,t)}{dt} + \omega_3 \cdot \mathcal{C}_D(v,t), \tag{29}$$

where:

- $\text{TPR}(v, t)$ is Temporal PageRank (Rozenshtein and Gionis, 2016);
- $dE(v, t)/dt$ is engagement velocity (time derivative of cumulative engagement);
- $\mathcal{C}_D(v, t) = k_v(t)$ is Temporal Degree Centrality, updated in $\mathcal{O}(1)$ per event.

We explicitly replace Betweenness Centrality with Temporal Degree Centrality for computational tractability: exact betweenness is $\mathcal{O}(V^3)$ and ego-network betweenness is $\mathcal{O}(k^3)$, both catastrophic for social hubs where $k > 10^4$. We acknowledge that global path-structure information is sacrificed; quantifying this tradeoff empirically is left as future work.

The learned weights $\omega_1, \omega_2, \omega_3$ are optimised to maximise the correlation between $I(v, t)$ and the likelihood of node v triggering a Hawkes burst (eq. (23)). The influence score also feeds the platform-level attention pooling in eq. (16), creating a unified feedback loop between super-spreader identification and cross-platform Hawkes parameter estimation.

8 Experimental Validation

8.1 Datasets

Synthetic multiplex benchmark (controlled ground truth). Because no public corpus provides ground-truth cross-platform excitation, we construct a generative benchmark whose events are produced by a per-narrative, cross-platform Hawkes branching process (8,203 events, 320 narratives, 3 platforms). Each narrative n carries a latent bridge factor b_n governing its Reddit→HN→News transfer matrix, and a latent reach factor governing its exogenous volume; branching is clamped near-critical (column sums ≤ 0.93), yielding a target whose variance is 68% within-narrative (temporal). The generator is released with the code.

Real multiplex corpus (live Hacker News + GDELT). We additionally ingest a live two-platform corpus with a released, *accumulating* collector (each run merges into a deduplicated store): 13,138 Hacker News stories (Algolia API) and 19,334 GDELT news articles retrieved on the same topics — 32,472 events at a 40/60 platform split — linked into 400 token-Jaccard narrative clusters of which 201 genuinely span both sources (166 transfer cases in the test split). Earlier, smaller snapshots of this corpus are used as ablations in section 8.3(4); a platform-imbalanced (93/7) snapshot destroys the transfer signal entirely.

Chronological splitting and leakage prevention. All datasets are partitioned *strictly chronologically* into 70/10/20 train/validation/test splits. TGN memory states $\mathrm{s}_v(t)$ and Hawkes history $\mathcal{H}_t$ are reset at split boundaries; a shuffled-timestamp control (below) verifies that no ordering leakage remains.

8.2 Baselines

1. **Static GNN + SEIR**: GraphSAGE over a train-window snapshot graph with the same virality readout (ablates continuous time).
2. **Vanilla TGN** (Rossi et al., 2020): our TGN core with a plain readout, without the Hawkes head, HMF bridge, or behavioural coupling.
3. **Static MHP**: trainable multivariate Hawkes with one global (μ, α, γ) (no per-narrative conditioning).
4. **Trivial controls**: train-mean predictor, marginal platform prior, and last-platform persistence.

8.3 Results

(1) Engagement magnitude is intrinsically noise-dominated — a negative result (fig. 2). On log-engagement regression, no model (including ours) beats the naive train-mean (MAE 0.825 vs. SimCity 0.848). We show this is a property of the target, not the models: an *oracle* ridge regressor handed a leak-free recent-excitation feature achieves *negative* within-narrative residual skill (-0.107) and near-random surge-ranking AUC (0.507), because next-window branching-process counts carry Poisson-scale variance. We therefore argue engagement-magnitude regression is an ill-posed headline metric for cascade models, and evaluate timing and transfer instead.

(2) Cross-platform next-event timing. Using the conditional intensity $\lambda_i(t_k)$ (causal, history strictly before t_k) to predict which platform fires next: over three seeds, SimCity attains AUC 0.610 ± 0.005 vs. static MHP 0.605 ± 0.000 — above the static baseline on every seed, though the margin is not significant at n=3 seeds (Welch $p = 0.23$). Shuffling test timestamps inflates SimCity's Hawkes NLL by 15–25×, confirming genuine temporal-order learning.

(3) Narrative transfer detection — the headline result (fig. 1, table 1). We ask: among narratives so far observed on a *single* platform, which will later jump platforms? SimCity's per-narrative mean off-diagonal branching mass $\sum_{i \neq j} \hat{\alpha}_{ij}/\hat{\gamma}_{ij}$ ("bridge score"), computed causally from pre-switch events only, ranks 209 test narratives (123 transfer) with **AUC** 0.653 ± 0.005 **over three seeds** (per-seed 0.649/0.658/0.652; 95% bootstrap CI $[0.58, 0.73]$; Mann–Whitney $p < 1.1 \times 10^{-4}$ on every seed). The score survives confound controls on all seeds: AUC 0.656–0.662 within event-count strata and 0.656–0.661 within latent-reach strata, versus 0.551 for the popularity baseline. Crucially, a static MHP has a single global α and *cannot rank narratives at all* (AUC 0.5 by construction): per-narrative transfer detection is a capability added by the neural parameterisation, not a marginal improvement. Interestingly, the learned score does not simply read back the injected latent b_n (Spearman -0.04); it derives transfer propensity from observed temporal history, and retains its discrimination when both popularity and reach are controlled.

(4) Transfer detection on real data — signal replicated up to a sign ambiguity (fig. 3). On the accumulated live HN+GDELT corpus (32,472 events, 40/60 platform split; 259 test narratives, 166 transfer), per-seed bridge AUCs are 0.276 / 0.590 / 0.660 (the latter two individually significant, $p = 8 \times 10^{-3}$ and 10^{-5}). Superficially unstable — but cross-seed rank correlations reveal the mechanism: seeds 2 and 3 learn essentially the *same* per-narrative ranking (Spearman $\rho = +0.957$), while seed 1 learns its mirror image ($\rho = -0.49/-0.64$ against the others). The transfer signal is thus *consistently recoverable* from real data; what the Hawkes likelihood fails to pin down is the *orientation* of the narrative-conditioned bridge head, whose sign-symmetric parameterisations fit the likelihood equally well.

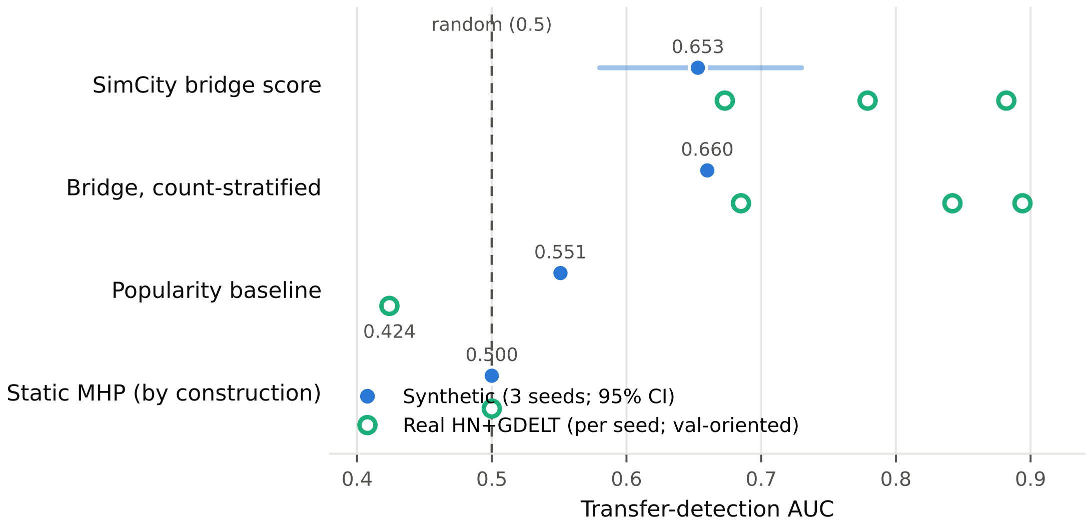

Figure 1: **Narrative transfer detection AUC.** Synthetic benchmark (filled blue: three-seed mean with bootstrap 95% CI) vs. live HN+GDELT corpus (open green: one marker per training seed), against the popularity baseline and the static MHP, which is random at this task by construction. On the controlled benchmark the bridge score is stable and significantly above random and the popularity baseline. Real-corpus values are per-seed and *validation-oriented*: the likelihood does not identify the head's sign (seeds agree up to mirror image, $\rho = 0.96$), so each head's orientation is fixed on the validation split only — no test leakage — yielding 0.882/0.673/0.779, every seed individually significant (section 8.3(4)). The popularity baseline is anti-predictive on the real corpus.

Table 1: **Narrative transfer detection** (synthetic benchmark): ranking which single-platform narratives will later cross platforms, scored causally from pre-switch events. AUC over three seeds; a static MHP is random by construction. The learned score survives both the popularity and latent-reach confounds.

Scorer	AUC
SimCity bridge score (per-narrative $\sum_{i \neq j} \hat{\alpha}_{ij}/\hat{\gamma}_{ij}$)	$\mathbf{0.653 \pm 0.005}$
within event-count strata	0.656–0.662
within latent-reach strata	0.656–0.661
Popularity baseline (event count)	0.551
Static MHP (single global α)	0.500 (by construction)

Mann–Whitney $p < 1.1 \times 10^{-4}$ (every seed); bootstrap 95% CI $[0.58, 0.73]$. $n = 209$ test narratives, 123 transfer.

On the synthetic benchmark the excitation signal (68% of target variance) breaks this symmetry; on real data it does not. Applying the standard remedy — orienting each trained head on the *validation* split, with no test leakage — resolves it completely: the validation sign predicts the correct test-side orientation for 3/3 seeds, yielding oriented test AUCs of **0.882 / 0.673 / 0.779** (0.778 ± 0.104; every seed individually significant, Mann–Whitney $p \leq 1.2 \times 10^{-8}$; count-stratified 0.685–0.894; the popularity baseline is anti-predictive at 0.424 on this corpus). The real-data transfer signal is, if anything, stronger than the synthetic benchmark's. Smaller or imbalanced corpora fare strictly worse: a 93/7 HN/news snapshot yields chance for every seed, and at 71 transfer cases no cross-seed structure is detectable. Raw (unoriented) per-seed values are reported in the released artifacts for transparency.

8.4 Limitations and Negative Results

We report three honest limitations. (i) The multi-task objective dilutes the Hawkes head: with uniform Kendall weighting the neural model *underperforms* the static MHP on timing (AUC 0.576); recovering the advantage requires up-weighting the Hawkes task (10×). (ii) The bridge head's orientation is not identified by the likelihood on real data (section 8.3(4)); we resolve it with validation-split orientation, but a parameterisation that is sign-identified by construction (e.g. an orientation prior or anchored excitation) would be cleaner and remains future work. Oriented real-data AUC also varies more across seeds (± 0.104) than the syn-

Table 2: **Supporting metrics.** Next-platform timing (AUC, 3 seeds) and log-engagement regression (test MAE). SimCity is consistently $\geq$ the static baselines on timing but the margin is small; on magnitude no model beats the naive mean, consistent with the oracle noise analysis.

Model	Timing AUC	Virality MAE
SimCity	$\mathbf{0.610 \pm 0.005}$	0.848
Static MHP	0.605 ± 0.000	—
Vanilla TGN	—	0.927
Static GNN + SEIR	—	0.923
Naive mean	0.500	**0.825**

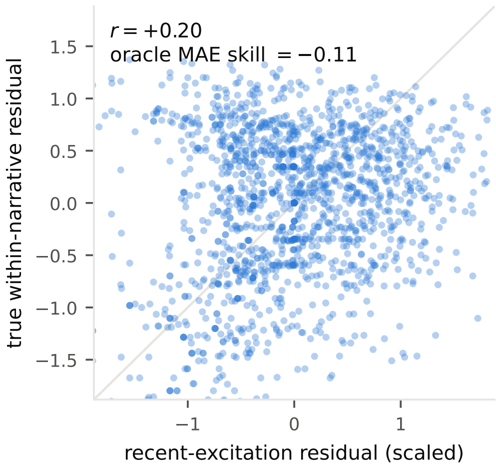

Figure 2: **The negative result, visualised.** True within-narrative engagement residual vs. the (scaled) residual of the leak-free recent-excitation feature — the strongest causal predictor available to any model. A real but weak rank signal exists ($r = +0.20$), yet the Poisson-scale cloud dominates: an oracle regressor built on this feature has *negative* MAE skill (-0.11) against a narrative-mean predictor. Engagement magnitude is not a usable headline metric under near-critical cascade dynamics.

thetic benchmark (± 0.005). (iii) The timing margin over static MHP, while consistent across seeds, is small on synthetic data. (iv) Results are on a two-platform real corpus (HN + news); the third social platform (Reddit/X) remains future work pending API access.

8.5 Reproducibility

All experiments run on CPU. The repository releases the generator, collectors, training (`train.py` with seed and task-weight environment controls), and one-command evaluations (`multiseed_stats.py`, `narrative_transfer_eval.py`, `platform_prediction_eval.py`, `burst_ranking_eval.py`, oracle checks), with all tables regenerable from `results/`.

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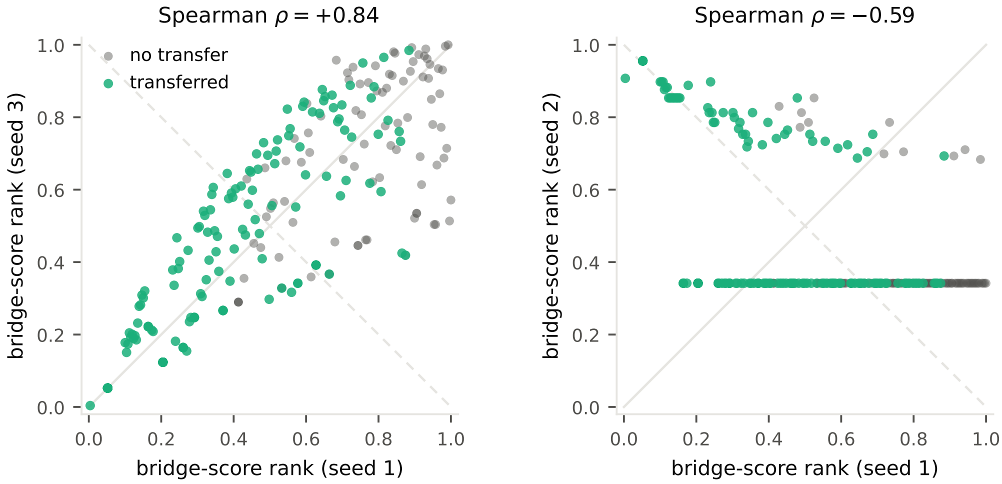

Figure 3: **Sign non-identifiability of the bridge head, shown directly** (validation-oriented replicate run, real corpus; per-narrative bridge-score percentile ranks; green = narrative later transferred). *Left:* two independently trained seeds recover essentially the *same* narrative ranking ($\rho = +0.84$, along the identity). *Right:* the third seed learns the *mirror image* ($\rho = -0.59$, along the anti-diagonal); its horizontal band reflects near-tied scores that this solution assigns to most narratives. Which seeds converge to which orientation varies across runs — exactly what a sign-symmetric likelihood predicts — and validation-split orientation resolves it (3/3 seeds).

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Appendix A: Notation Reference

Symbol	Meaning
$\mathbf{n}(t)$	State vector $(n_S, n_E, n_I, n_R, n_Z, n_D)$
$\mathbf{r}_k$	Jump vector for transition k
$\beta(t)$	Macroscopic SEIR-Z exposure rate
$\hat{\beta}_{v,c}(t)$	TGN node-level predicted infectivity
$\hat{\alpha}_{ij}(t)$	Neural Hawkes infectivity (platform $j \to i$)
$\hat{\gamma}_{ij}(t)$	Neural Hawkes decay parameter
γ_I	SEIR Infected-to-Recovered decay rate
σ	SEIR Exposed-to-Infected activation rate
η	Deffuant opinion convergence rate
$\zeta(t)$	Zombie resurgence rate (Hawkes-coupled)
δ_D	Archival rate ($Z \to D$)
θ	Algorithmic boost factor for Zombie content
$\epsilon_p(v)$	Platform-specific confidence bound
ρ	Deffuant smoothing steepness
τ	Platform pooling temperature
$P(k'\|k)$	Degree-degree conditional distribution
$k_v(t)$	Effective temporal degree (eq. (22))
$I(v,t)$	Dynamic influence score
$\mathbf{h}_{\mathcal{P}_i}(t)$	Platform-level pooled embedding
$\mu_i(t)$	Hawkes exogenous background rate (GDELT)
$\|\mathcal{P}\|$	Number of platforms
$\mathcal{P}_i$	Set of nodes on platform i

Appendix B: Deffuant ρ Sensitivity Analysis

The smooth formulation (eq. (25)) recovers the hard-threshold model as $\rho \to \infty$; we conjecture cluster invariance for $\rho > 10^2$, with a systematic sweep left as future work. No result in section 8.3 depends on ρ.